

of IT and OT marks a turning point in the evolution of industries.

Given the competitive landscape, private wireless is emerging as a dominant force. “Globally, there are roughly 7 million macro base stations, essentially communication towers. However, when you look at the industrial enterprise or the industrial space, the interesting fact is that globally, there are 14 million industrial sites,” says Abin Chacko Moozhiyil, Global Marketing Practice Leader, Network & Edge, Kyndryl. Industries ranging from manufacturing and warehousing to healthcare are all vying to tap into the advantages of private wireless. Nevertheless, a one-size-fits-all approach won’t suffice. Each industry, with its unique operational intricacies, demands custom-tailored private wireless solutions.

Wireless’ evolution to Wi-Fi 6: Do we really need private wireless?

With the tech world constantly evolving, businesses and consumers are often inundated with various options to meet their connectivity needs. One prominent question that continues to resonate is: ‘With Wi-Fi technology, especially the evolving Wi-Fi 6, why is there an urge to explore private wireless?’ To shed light on this, we dive into private wireless technology’s advantages and potential applications.

Mission-critical applications.

Wi-Fi is undoubtedly efficient in standard IT environments, seamlessly connecting employees, contractors, and systems. However, for mission-critical applications demanding ultra-low latency, Wi-Fi exhibits limitations in scenarios where every millisecond counts. Wi-Fi often falls short, making private wireless a more viable solution.

Coverage superiority. Wi-Fi’s prowess diminishes when faced with challenges like thick concrete walls

Flexibility and cost-efficiency

Private wireless networks, including private LTE and 5G, are proving to be effective alternatives to traditional wired connections, particularly in scenarios where such connections are not feasible. Consider these key points:

- They can be deployed flexibly, providing reliable connectivity in large, expansive environments.
- These networks reduce the substantial costs of installing and maintaining large-scale wired infrastructures.
- They are subject to ongoing improvements, with advancements anticipated in roaming and mobility.

Despite these technologies still being in their development phases, they offer promising prospects for the future. Private wireless networks are poised to play a significant role in the connectivity landscape by facilitating widespread and reliable connectivity across various contexts. With their ability to provide cost-effective and adaptable solutions, they are worth watching as we navigate towards a more connected future.

or isolated environments typical in industrial settings. In contrast, private LTE technology allows for three extra wall penetrations. This difference in reach can be crucial in various industrial applications.

Device connectivity. Another advantage of private wireless is its capability to support a larger number of devices. While Wi-Fi might be limited to connecting about 16 devices, private networks based on LTE or 4.9G can support up to 840 active connected devices. As the world leans heavily towards IoT, this difference in connectivity can significantly impact operational efficiency.

Cost efficiency. While the initial perception suggests Wi-Fi is the cheaper alternative, a deeper analysis reveals a different story. Considering factors like latency, the number of connected devices, and long-term maintenance, private wireless often emerges as the more economical choice. “Studies reveal that private wireless networks are up to 20% more economical than typical Wi-Fi networks,” says Abin.

Industrial applications. Private wireless is carving out a unique presence across multiple industrial domains. Manufacturing, for instance, is spearheading its adoption. Energy sectors, utilities, and even unmanned aerial vehicles (UAVs) for surveil-

lance find the connectivity of private wireless more suitable than Wi-Fi. In the healthcare industry also, where efficient, uninterrupted connectivity can be life-saving, the shift towards private wireless solutions is quite evident.

As technological advancements in private wireless networks continue to reshape industries and optimise operations, certain partnerships stand out for their potential impact on a nation’s infrastructure. “Recently, Nokia and the Adani Group announced plans to deploy private wireless technology at various airports and seaports in India that they oversee and manage,” Abin adds. Additionally, private wireless offers hope in education, especially in areas grappling with connectivity challenges, bridging the gap and ensuring continuous learning.

Private wireless implementation in real-world scenarios

The rapid evolution of the digital landscape has positioned private wireless technology as a pivotal solution for many industrial sectors. Presented here are two distinct case studies demonstrating the transformative potential of private wireless solutions in addressing real-world manufacturing challenges.

Leading metal producer in India. An Indian conglomerate, a titan in